ZCAS University

SCHOOL OF COMPUTING TECHNOLOGY AND APPLIED SCIENCES

CCS3142 MOBILE APPLICATION DEVELOPMENT

FINAL SEMESTER EXAMINATION

TUESDAY, 5 DECEMBER, 2023

08:30 - 11:30 HOURS

TIME ALLOWED : WRITING – THREE HOURS : READING – 5 MINUTES

INSTRUCTIONS:

1. Section A: this question is **compulsory** and must be attempted.
2. Sections B: Answer **THREE (3)** questions from this section.
3. This examination paper carries a total of **100 marks**.
4. Candidates must **not turn this page** until the invigilator tells them to do so.

SECTION A: Question 1 is compulsory and must be attempted

NOTE: Your answers should NOT be brief. Give detailed essay style reasoned answers with structured paragraphs with headings.

Question 1

Emulators, pivotal in the Android Software Development Kit (SDK), simulate device behaviors crucial for app testing and debugging. Operating within the Android system architecture, emulators engage with the Hardware Abstraction Layer (HAL), harmonizing with layered elements like native libraries, Java APIs, and core system services. This cohesive structure converges within the Android development environment, spearheaded by tools such as Android Studio. This ecosystem nurtures innovation by providing developers a comprehensive platform to ideate, create, and refine applications. Seamlessly integrating emulators, SDK functionalities, and system architecture, this environment propels the dynamic landscape of Android app development, empowering developers to craft cutting-edge solutions for an evolving mobile ecosystem.

a) The Android *Emulator* is a software tool that simulates an Android device on a computer. Describe in detail any *five* features of the emulator. **(10 marks)**

b) The *Android SDK* (Software Development Kit) is a collection of software tools and resources provided by Google for developers to create Android applications. Identify and explain any *five* components of the SDK. **(10 marks)**

c) The Android operating *system architecture* is designed in a layered structure, comprising multiple components that interact with each other to provide a robust environment for running applications. Explain the roles and interactions among at least *three* major layers, highlighting how they contribute to the overall functionality of the system. **(10 marks)**

d) Outline the step-by-step process for setting up the Android *development environment* on a Windows, linux or macOS system. Include details about downloading and installing necessary software, configuring SDK tools, and ensuring proper integration with Android Studio for seamless app development. **(10 marks)**

[Total: 40 Marks]

SECTION B: Attempt any THREE questions in this section

Question 2

a) Discuss the rational of categorization of mobile apps based on their usage context or target audience by highlighting at least three categories. **(6 marks)**

b) Discuss at least three core principles that differentiate microservices architectures from monolithic architectures. **(6 marks)**

c) Study the code snipets shown in the table below. Explain the purpose of each widget and describe the practical use cases. **(8 marks)**

TextField(
  decoration: InputDecoration(
  labelText: 'Enter text',
  border: OutlineInputBorder(),
  ),
  onChanged: (value) {
  // Handle text changes here
  },
)
	bool checkBoxValue = false;
	
	Checkbox(
	  value: checkBoxValue,
	  onChanged: (value) {
	  setState(() {
	  checkBoxValue = value!;
	  });
	  },
	)

ElevatedButton(
  onPressed: () {
  // Add functionality here
  },
  child: Text('Elevated Button'),
)
	RadioListTile(
	  value: 1,
	  groupValue: selectedRadio,
	  onChanged: (value) {
	  setState(() {
	  selectedRadio = value;
	  });
	  },
	  title: Text('Radio Button 1'),
	  ),

[Total: 20 Marks]

Question 3

a) Code obfuscation refers to the process of transforming code to make it more challenging to understand or reverse-engineer while preserving its functionality. Evaluate three benefits and limitations of code obfuscation in mobile application software development. **(6 marks)**

b) Persistence storage in software development refers to the technique of storing data permanently. Explain the *three* persistence storage mechanisms available in Android development for data retention. **(6 marks)**

c) Design a user preference settings screen using Flutter that allows users to customize their app settings. The interface should include the following input and display widgets:
Text: Display a heading at the top of the screen such as "User Preferences."
TextFields: Create two text fields for users to input their first name and last name.
(8 marks)

[Total: 20 Marks]

Question 4

a) Below are widgets that are commonly used in mobile application interfaces. Write Flutter dart code segment outlines that will create these widgets.

1. Text
2. TextButton
3. Icon
4. Children **(8 marks)**

b) Explain the fundamental concepts of animation in Flutter. Define terms like AnimationController, Tween, and AnimationBuilder. Describe how these components work together to create smooth and interactive animations within a Flutter application.
(6 marks)

c) Below is an image that depicts a user authentication screen. Write flutter code that will generate the screen below. **(6 marks)**



[Total: 20 Marks]

Question 5

a) App resource constraints refer to limitations and boundaries within which mobile applications must operate due to inherent device capabilities, system restrictions, and user experience considerations. Identify and evaluate the impact of any *three* constraints on app performance. **(6 marks)**

b) In the realm of software development, plugins refer to modular components or extensions that enhance the functionality of an existing application or framework. Discuss at least three key benefits that plugins offer to developers when building mobile applications. **(6 marks)**

c) Study the code below. Explain the purpose and discuss the importance of its significance in user interface functionality. **(8 marks)**

```
import 'package:flutter/material.dart';

void main() {
  runApp(MyApp());
}
```

```
class MyApp extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      home: MyForm(),
    );
  }
}

class MyForm extends StatefulWidget {
  @override
  _MyFormState createState() => _MyFormState();
}

class _MyFormState extends State<MyForm> {
  final GlobalKey<FormState> _formKey = GlobalKey<FormState>();
  String _name = '';

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: Text('Form Validation Example'),
      ),
      body: Padding(
        padding: EdgeInsets.all(16.0),
        child: Form(
          key: _formKey,
          child: Column(
            crossAxisAlignment: CrossAxisAlignment.start,
            children: <Widget>[
              TextFormField(
                validator: (value) {
                  if (value == null || value.isEmpty) {
                    return 'Please enter your name';
                  }
                  return null;
                },
                onChanged: (value) {
                  setState(() {
                    _name = value;
                  });
                },
                decoration: InputDecoration(
                  labelText: 'Name',
                ),
              ),
              SizedBox(height: 20),
              ElevatedButton(
                onPressed: () {
```

```
              if (_formKey.currentState!.validate()) {
                // If the form is valid, proceed with desired actions
                ScaffoldMessenger.of(context).showSnackBar(
                  SnackBar(content: Text('Form is valid')),
                );
              }
            },
            child: Text('Submit'),
          ),
        ],
      ),
    ),
  ),
 );
 }
}
```

[Total: 20 Marks]

END OF EXAMINATION